

Nexus™ Sovereign AI Workforce

Dual Executive & Engineering Guidebook

A comprehensive operational manual for managing 18 autonomous agents, live digital vending machines, automated cashflow gateways, and self-healing cloud serverless functions.

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Location: Grand Baie, Mauritius

Public Gateway: <https://nexusbots-nu.vercel.app>

Local Headquarters: <http://127.0.0.1:8000>

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Level 0: The Quick-Start Story (For Beginners & 10-Year-Olds)

Imagine you own a **secret digital treehouse with 18 loyal robot friends**. You are the CEO sitting in the comfortable chair, and your robots never sleep.

How the Vending Machine Works in 4 Steps:

1. **Scout:** A robot checks YouTube to see what 500,000 coders are struggling with.
2. **Build:** Another robot writes a clean helper program in 2 seconds and tests it in a safe test-tube.
3. **Sell:** It places the tool on your online shelf for **\$1.00 USD (or Rs 45 MUR)** with a 1-click buy button.
4. **Earn:** When someone in London or Tokyo buys it, PayPal or MCB Juice takes the cash and delivers the file instantly.

Part 1: The CEO Executive Operating Manual

As Chief Executive Officer, your focus is on **cash flow, risk mitigation, product throughput, and system health**. You do not need to write code. Your daily commitment is approximately **3 minutes**.

1. Cash Flow & Automated Financial Pipelines

Nexus operates two independent payment collection arteries:

Artery	Target Audience	Price Point	Settlement & Verification
PayPal REST Gateway	International developers (US, Europe, Asia)	\$1.00 USD one-time	Automated instant capture via OAuth2 bearer token. Direct deposit into your linked account.
MCB Juice Portal	Mauritius domestic clients & SME owners	Rs 45.00 MUR one-time	Direct mobile transfer to +230 58169420 . Verified against replay fraud via <code>financial_shield</code> .

2. The 24/7 Autopilot Night Shift

Every 30 minutes while you sleep, the **Night Shift Chronicle** runs an automated operational sweep:

- **Inbox Hygiene:** Automatically incinerates marketing spam and cold pitches across all 5 configured IMAP accounts.
- **Zero Password Deletions:** Banking alerts, 2FA codes, and OTPs are protected by the Immunity Shield.
- **Infrastructure Cap:** Monitors cloud burn to guarantee expenses stay well below the **\$180 USD monthly budget**.

- **Morning Standup Briefing:** When you wake up, your Executive AI Partner has a single markdown summary ready on your screen.

3. The 3 CEO Dashboard Levers

Widget	Where to Look	What It Tells You
Executive Ledger	Top-Right Card	Cryptographically signed record of every rupee and dollar earned.
Fleet Health Matrix	Center Grid	18 green status indicators proving all employee agents are standing by.
1-Click Factory Box	Bottom Toolbar	Type any keyword (e.g., <i>"invoice parser"</i>) to manufacture a new live product.

Part 2: The Engineering Technical Manual (In Plain English)

Nexus is architected as a **hybrid edge-and-core sovereign system**. It combines high-speed global delivery with deep local privacy.

```
[ GLOBAL PUBLIC INTERNET ]           [ LOCAL MACHINE / SOVEREIGN CORE ]
https://nexusbots-nu.vercel.app      http://127.0.0.1:8000
|                                     |
└─▶ Vercel Serverless Gateway (api/index.py)   └─▶ FastAPI Master Engine
(server.py)                                     |
└─▶ Static Glassmorphic Store (/store)         └─▶ 18 Autonomous Agent Loops
└─▶ Dynamic Tools REST API (/api/tools)         └─▶ Live IMAP SSL Sync
(Gmail/Outlook)                               |
└─▶ Read-Only /tmp Storage Layer               └─▶ Tamper-Resilient Signed
Ledgers
```

1. The 9 Universal Agent Tools

All agents have access to the Universal Tool Registry (core/tool_registry.py):

Tool Name	Category	Technical Function
niche_scout	Market Scout	Searches public developer discussions for recurring technical pain points.
scout_video_trends	Market Scout	Parses YouTube video view counts to quantify real consumer demand.
verify_email_domain	Lead Generation	Performs DNS socket MX record resolution to prevent cold email bounces.
generate_whatsapp_link	Communication	Creates dynamic wa.me deep-links with URL-encoded bilingual sales pitches.
create_paypal_link	Finance	Calls PayPal v2 Orders API to generate 1-click checkout URLs with capture tokens.
check_paypal_balance	Finance	Queries official PayPal Reporting API via OAuth2 client credentials.
run_safe_diagnostic	System	Verifies localhost port availability, CPU load, and network latency.
run_python_sandbox	QA / Security	Compiles bytecode and runs scripts in isolated subprocesses with 6s timeouts.
broadcast_product_announcement	Marketing	Dispatches formatted embed announcement cards to Discord/Slack webhooks.

2. The E2B Isolated Code Sandbox

Nexus guarantees that no customer receives broken code. The **Sandbox Executor** (`core/sandbox_executor.py`) conducts a strict two-phase inspection:

1. **Static Compilation (Bytecode):** Runs `py_compile` to catch missing brackets, invalid syntax, or unclosed strings.
2. **Subprocess Execution:** Spawns an isolated Python runtime with cross-platform UTF-8 encoding and a strict 6,000ms execution cap.
3. **Automatic Quarantine:** Any script that produces an uncaught exception is withheld from the store catalog until auto-corrected.

3. MemGPT 3-Tiered Memory Architecture

- **Working Memory:** In-memory dictionary tracking current session goals and transient tasks.
- **Recall Memory (`memory/recall_memory.json`):** Sliding window of the last 100 scouted niches, tool executions, and client responses.
- **Archival Memory (`memory/archival_memory.json`):** Immutable corporate ground truth (owner information, MCB Juice phone numbers, verified product blueprints).

4. Serverless Read-Only Safeguard (Vercel Architecture)

Vercel serverless functions run on AWS Lambda where the application directory (`/var/task`) is strictly read-only. The storage subsystem (`core/storage.py`) transparently redirects state writes to `/tmp` whenever `VERCEL=1` is detected, preventing filesystem crash errors and ensuring 100% cloud uptime.

Nexus™ Autonomous Intelligence Workforce

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